**QP CODE: 25900219**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025**

**SECOND SEMESTER**

**Discipline Specific Core Course (DSC) - MG2DSCZGY101 - BIOLOGICAL BASIS OF BEHAVIOR II**

*(2024 Admission onwards)*

**Duration: 1.5 Hours** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

**Part A**

Multiple Choice Questions, Answer all questions

Each question carries **1** marks

1. Which structure is found in both prokaryote and eukaryote? [K] / [CO1]
   Ans: (c) ribosome

2. During which phase of the cell cycle does DNA replication occurs? [K] / [CO1]
   Ans: ( a) S-phase

3. Endoplasmic reticulum with ribosomes are called. [K] / [CO1]
   Ans: (b) Rough endoplasmic reticulum

4. Why are sex-linked traits more common in males? [U] / [CO2]
   Ans: (b) Males have only one X chromosome

5. What is the function of a gene? [K] / [CO2]
   Ans: (a) To control cell division To carry genetic information or ( b) to determine trait (can give full mark for both options)

6. Which of the following is a pyrimidine base in DNA? [K] / [CO2]
   Ans: (a) cytosine or (d) Thymine (can give full mark for both options)

7. What is a sex-limited trait? [K] / [CO2]
   Ans: (b) expressed only in one sex

8. Why do people with Down syndrome have intellectual disabilities? [U] / [CO3]
   Ans: (a) extra chromosome 21

9. Euthenics focuses on improving the quality of life through what? [U] / [CO4]
   Ans: (b) better nutrition and living conditions

10. Which of the following is NOT a common reason for seeking genetic counselling? [An] / [CO4]

Ans: (d) general healthcare checkup **[1x10 = 10]**

## Part B

Short Answer Questions, Answer any **10** questions

Each question carries **2** marks

11. What is cytokinesis? [K] / [CO1]

Cytokinesis is the process that follows nuclear division, where the cytoplasm of a parent cell divides into two daughter cells, each with its own nucleus and cellular components ( definition- 2 mark).

12. Explain the role of neurotransmitters in cell signalling. [U] / [CO1]

Neurotransmitters are chemical messengers that transmit signals across synapses from one neuron to another, enabling communication within the nervous system. They bind to receptors on target cells, triggering a cascade of events that can either excite or inhibit the cell's activity, regulating various physiological processes.

Transmission of Nerve Impulses, Regulation of Mood and Behavior, muscle contraction, Modulation of Physiological Functions, Memory and Learning, Regulation of Pain (any 2 roles)

13. Define meiosis. [U] / [CO1]

Meiosis is a type of cell division that reduces the chromosome number by half, producing four haploid cells (sperm or eggs). It occurs in two stages: meiosis I (separates homologous chromosomes) and meiosis II (separates sister chromatids), ensuring genetic diversity in offspring. (definition - 2 mark)

14. What is monohybrid cross? Give an example. [U] / [CO2]

A monohybrid cross is a genetic cross that examines the inheritance of one trait. Eg: flower colour in pea plants, seed colour in pea plants etc. ( definition 1 mark, one example 1 mark )

15. Which are the different sites in a ribosome? [U] / [CO2]

A site (Aminoacyl site): Where the incoming tRNA carrying an amino acid binds.

P site (Peptidyl site): Holds the tRNA with the growing polypeptide chain.

E site (Exit site): Where the empty tRNA exits the ribosome after delivering its amino acid. (mention the three sites - 2 mark).

16. Define a reciprocal cross. How does it differ from a normal cross? [U] / [CO2]

A reciprocal cross is a genetic cross where the sexes of the parents are reversed to observe if the inheritance pattern of a trait is sex-dependent.

Difference from a normal cross: In a normal cross, the traits are inherited based on a specific male and female parent arrangement, while in a reciprocal cross, the male and female roles are switched to examine potential differences in inheritance patterns. (definition of reciprocal cross - 1 mark, difference - 1 mark OR Explanation with example 2 mark.

17. How many types of gametes are produced in a dihybrid cross? Describe with suitable example. [U] / [CO2]

In a dihybrid cross, four types of gametes are produced in heterozygous condition. For eg: Parental Genotypes: RrYy - Possible Gametes RY, Ry, rY, ry

Homozygous condition RRYY – Possible gamete RY

rryy –Possible gamete – ry

(no. of gametes- 1 mark, example: 1 mark) (Any one condition can give full mark)

18. Explain the chromosomal makeup of a person with Klinefelter's syndrome. [U] / [CO3]

**Klinefelter's syndrome**: 47, XXY. A normal male has 46 chromosomes, with one X and one Y (46, XY), while an individual with Klinefelter's syndrome has 47 chromosomes, with two X chromosomes and one Y (47, XXY) along with 44 autosomes. (2 mark)

19. What is the role of a karyotype in detecting chromosomal abnormalities? [K] / [CO4]

A **karyotype** helps in detecting chromosomal abnormalities by: Finding extra or missing chromosomes (e.g., Down syndrome), Identifying chromosome changes (e.g., deletions or rearrangements), Diagnosing genetic conditions (e.g., Klinefelter's or Turner syndrome). (Any 2 role: 1 mark each)

20. Explain somatic mutation. [U] / [CO3]

A **somatic mutation** is a change in the DNA sequence that occurs in **somatic cells** (any cells in the body that are not reproductive cells). These mutations are not inherited but can affect the individual's health, potentially leading to diseases like cancer. (definition: 2 marks).

21. What is frameshift mutation? [U] / [CO3]

A **frameshift mutation** happens when nucleotides are added or deleted in the DNA, shifting the reading frame of the genetic code. This changes the sequence of amino acids in the protein, usually making it nonfunctional. It can lead to serious diseases like cancer or genetic disorders. (define - 2 mark).

22. Explain the role of selective breeding in eugenics. [U] / [CO4]

Selective breeding is the process of choosing specific plants or animals with desirable traits to reproduce, in order to pass those traits on to the next generation**. Improve traits**: Pass on good traits like health or intelligence /**Control genetics**: Choose who reproduces to shape future generations /**Remove bad traits**: Stop passing on harmful traits /**Shape population**: Change the genetic makeup of a group /**Ethical concerns**: Can lead to discrimination and loss of rights. (definition: 1 mark, any 2 roles: 1 mark)

**[2x10 = 20]**

## Part C

Short Essay Questions, Answer any **5** questions

Each question carries **4** marks

23. What is Cell Communication, and Why is It Important for Living Organisms? [U] / [CO1]

**Cell communication** is the process by which cells send and receive signals to each other, allowing them to coordinate activities, respond to changes, and function properly. This can happen through chemical signals, like hormones, or physical signals, like direct contact between cells. **Roles**: Coordination, Response to Environment, Growth and Development, Maintaining Homeostasis, Immune Response, Cell Repair and Maintenance, Regulation of Metabolism, Tissue Formation, Reproduction, Prevention of Diseases. (definition: 1 mark, any 3 roles with explanation or any 6 roles: 3 mark).

24. Define juxtacrine signalling. Give an example. [K] / [CO1]

**Juxtacrine signaling** is a type of cell communication where signaling molecules are passed directly between two adjacent cells through their membranes, typically via cell surface proteins. This requires physical contact between the signaling cell and the receiving cell. Any one example ( definition : 2 mark, example 2 mark).

25. Briefly explain transcription. [U] / [CO2]

**Transcription** is the process by which an **RNA molecule** is synthesized from a **DNA template**. During transcription, the enzyme **RNA polymerase** reads the DNA sequence of a gene and creates a complementary RNA strand. This RNA strand is called **messenger RNA (mRNA)**, which then carries the genetic information from the DNA in the nucleus to the ribosome, where proteins are made. Steps : initiation, elongation, termination. (Brief account)

26. Explain Law of Independent Assortment with an example. [U] / [CO2]

The **Law of Independent Assortment** states that genes for different traits are inherited independently of each other, meaning the inheritance of one gene does not affect the inheritance of another gene. Any dihybrid cross as an example. Definition 1.5 mark. Example : 2.5 mark

27. Discuss the experimental evidence that led to the acceptance of the Chromosome Theory. [U] / [CO2]

The Sutton and Boveri Theory, also known as the Chromosomal Theory of Inheritance, was proposed by Walter Sutton and Theodor Boveri in the early 1900s. Genes are located on chromosomes, and these chromosomes are responsible for passing traits from parents to offspring. Chromosomes come in pairs – just like Mendel's factors (genes). Chromosomes separate during meiosis, like genes do in Mendel's laws. Each parent gives one set of chromosomes (and genes) to their child. (Liberal valuation)

28. What is the role of gene mutations in chromosomal aberrations? [K] / [CO3]

**Gene mutation** is a **change in the DNA sequence** of a gene (1 mark).

A chromosomal aberration is a structural or numerical change in chromosomes (1 mark). Gene mutations can disrupt normal DNA repair and replication, increasing the risk of chromosomal aberrations. Mutations in genes regulating chromosome structure can lead to deletions, duplications, inversions, or translocations. (Any one effect of gene mutation- 2 marks)

29. How can pedigree analysis be used to evaluate the ethical considerations of genetic counseling for a family with a history of haemophilia? [An] / [CO4]

**Haemophilia** is a genetic bleeding disorder in which the blood does not clot properly due to the absence or low levels of certain clotting factors. It is usually inherited in an **X-linked recessive** pattern, meaning it primarily affects males, while females are typically carriers. Pedigree analysis helps trace how haemophilia is passed in a family. Pedigree analysis helps identify carriers and affected individuals, enabling family members to make informed choices about reproduction, prenatal testing, or early treatment options. It is useful in genetic counselling to: Find out who may carry or get the disease, help families make informed choices about having children, keep family information private, avoid pressure—counselors only guide, not force decisions, Prevent harm by planning ahead for care or treatment. By clarifying genetic risks through analysis, counselors can help prevent misconceptions and reduce the risk of social stigma or genetic discrimination within families or communities. (Brief Description - Any four relevant points- 4 marks)

**[4x5 = 20 Marks]**